Multiple Choice:

1. A cylindrical pressure vessel is fabricated from steel plates which have a thickness of 20 mm. The diameter of the pressure vessel is 500 mm and its length is 3 m. Determine the maximum internal pressure which can be applied if the stress in the steel is limited to 140 MPa.

   A. 1.12 MPa C. 3.32 MPa
   B. 2.42 MPa D. 3.68 MPa

2. A steel wire 10 m long and hanging vertically supports a tensile load of 2000 N. Neglecting the weight of the wire, determine the required diameter if the stress is not to exceed 140 MPa and the total elongation is not to exceed 5 mm.

   A. 4 mm C. 6 mm
   B. 5 mm D. 7 mm

3. What is the minimum diameter of a solid steel shaft that will not twist through more than 3° in a 6 m length when subjected to a torque of 14 kNm? G = 83 GPa.

   A. 102 mm C. 118 mm
   B. 110 mm D. 135 mm

4. A spring is made by turning a 15-mm diameter wire 25 times. If the radius of the turn is 100 mm, by how much will the spring elongate if a load of 1.5 kN is applied? Assume that G = 83 GPa.

   A. 571 mm C. 604 mm
   B. 592 mm D. 622 mm

5. The rails of a railroad are made of steel. They are laid abutting each other at their ends so that there is no room for expansion. Find the stress in the rails for a temperature change of 20ºC. For steel, coefficient of thermal expansion = 11.7 x 10^{-6} m/ºC and modulus of elasticity = 200,000 MPa.

   A. 44 MPa C. 52 MPa
   B. 47 MPa D. 56 MPa

6. A 100-m steel tape is 1.5 mm too short. If the tape's effective width is 10 mm and thickness of 0.11 mm, find the tensile pull necessary at each end of the tape for it to measure a distance of 100 m. Modulus of elasticity of steel = 200,000 MPa.

   A. 2.9 N C. 3.3 N
   B. 3.1 N D. 3.8 N

7. A cylinder has a mean radius of 100 mm and a thickness of 10 mm and an internal pressure of 5 MPa. What is the hoop stress of this cylinder?

   A. 40 MPa C. 50 MPa
   B. 45 MPa D. 55 MPa

8. A cast-iron column supports an axial compressive load of 250 kN. Determine the inside diameter of the column if its outside diameter is 200 mm and the limiting compressive stress is 50 MPa.

   A. 142 mm C. 168 mm
   B. 158 mm D. 183 mm

9. Solve for the diameter of a 10-meter solid steel rod if it should not twist more than 4° after subjecting it to a torque of 18 kN-m. Assume that the modulus of rigidity of steel is 83 GPa.

   A. 35 mm C. 42 mm
   B. 38 mm D. 45 mm

10. A 3-meter solid steel shaft, with an unknown diameter, twists by 5 degrees as it is stressed by 60 MPa. What is the diameter of the solid steel shaft? Assume that G = 83 GPa.

    A. 40 mm C. 46 mm
    B. 43 mm D. 50 mm

11. What is the maximum elongation of a helical spring which is made of phosphor bronze composed of 20 turns of 20 mm diameter wire with a mean radius of 80 mm when the spring is stressed to 140 MPa, for which G = 42 GPa?

    A. 226 mm C. 242 mm
    B. 235 mm D. 250 mm

12. A 3-meter solid steel shaft, with an unknown diameter, twists by 5 degrees as it is stressed by 60 MPa. What is the diameter of the solid steel shaft? Assume that G = 83 GPa.

    A. 41.2 mm C. 46.1 mm
    B. 43.5 mm D. 49.7 mm

13. A 20-m rod is suspended vertically from a ceiling. It has a cross section diameter of 300 mm and supports a tensile load of 10 kN. Find the total elongation if this rod is made of steel and has a unit mass of 7850 kg/m^3.

    A. 0.081 mm C. 0.096 mm
    B. 0.092 mm D. 0.105 mm

14. Solve for the minimum diameter of a solid steel shaft that will not twist through more than 3 degrees in a 6 m length if it is subjected to a torque of 12 kNm.

    A. 114 mm C. 125 mm
    B. 120 mm D. 138 mm

15. A steel column that has a diameter of 50 mm and 2 m long is surrounded by an iron shell that is 5 mm thick. What compressive load will shorten the combined system a total of 0.8 mm. For steel, E = 200 GPa. For iron, E = 100 GPa.

    A. 150 kN C. 185 kN
    B. 179 kN D. 192 kN

16. A torque of 1 kN-m is applied to a 40 mm diameter rod of 3 m length. Determine the maximum shearing stress induced. Take G=80 GPa.

    A. 5.38 x 10^7 N/m^2
    B. 6.24 x 10^7 N/m^2
    C. 7.96 x 10^7 N/m^2
    D. 4.56 x 10^7 N/m^2

17. A torque of 1 kN-m is applied to a 40 mm diameter rod of 3 m length. Determine the maximum twist produced. Take G=80 GPa.

A. 8.56° C. 6.24°
B. 6.40° D. 7.67°

18. A solid shaft of mild steel 200 mm in diameter is to be replaced by hollow shaft of alloy steel for which the allowable shear stress is 22% greater: If the power to be transmitted is to be increased by 20% and the speed of rotation increased by 6%, determine the maximum internal diameter of the hollow shaft. The external diameter of the hollow shaft is to be 200 mm.

A. 145 mm C. 178 mm
B. 104 mm D. 183 mm

19. 20 kNm torque is applied to a shaft of 7 cm diameter. Calculate the maximum shear stress in the shaft.

A. 297 MPa C. 201 MPa
B. 314 MPa D. 310 MPa

20. 20 kN-m torque is applied to a shaft of 7 cm diameter. Calculate angle of twist per unit length of shaft. Take 10^5 MPa.

A. 4.17° C. 6.24°
B. 3.82° D. 4.86°

21. A bar is suspended from a support and has an axial load of 10 kN. If the bar has a cross section of 20 mm^2, what is the stress experienced by the bar?

A. 500 MPa C. 560 MPa
B. 535 MPa D. 600 MPa

22. A bar is suspended from a support and has an axial load of 10 kN. If the bar has a cross section of 20 mm^2, what is the stress experienced by the bar?

A. 500 MPa C. 560 MPa
B. 535 MPa D. 600 MPa

23. Choose the wrong statement

A. Elongation produced in a rod (by its own weight) which is rigidly fixed at the upper end and hanging is equal to that produced by a load half its weight applied at the end.
B. The stress at any section of a rod on account its own weight is directly proportional to the distance of the section from the lower end.
C. Modulus of elasticity is having the same unit as stress.
D. If a material expands freely due to heating, it will develop thermal stresses.

24. If T = rise in temperature and α = co-efficient of linear expansion, which of the following gives the thermal stress?

A. $E\alpha T$ C. $\frac{ET}{a}$
B. $\frac{Ea}{T}$ D. $\frac{1}{EaT}$

25. A steel punch can be worked to a compressive stress of 800 MPa. Find the least diameter of hole which can be punched through a steel plate 10 mm thick if its ultimate shear strength is 350 MPa.

A. 45.2 mm C. 17.5 mm
B. 34.2 mm D. 28.3 mm

END